

Institute of Actuaries of India

Subject SP6 – Financial Derivatives Principles

June 2019 Examination

INDICATIVE SOLUTION

Introduction

The indicative solution has been written by the Examiners with the aim of helping candidates. The solutions given are only indicative. It is realized that there could be other points as valid answers and examiner have given credit for any alternative approach or interpretation which they consider to be reasonable.

Solution 1:**i) Discounting assets and liabilities we get**

	PV	1359.333	1362.885		3930.196	2700.225
Rate		liabilities	Asset	Discount	Wt L	WT A
10%	1	200	400	0.909091	200	400
10%	2	400	800	0.826446	800	1600
10%	3	400	450	0.751315	1200	1350
10%	4	800		0.683013	3200	0

	DMT	Modified duration
Liabilities	2.891267	2.628425
Assets	1.981257	1.801143

ii)

Change in assets	74	Fall in interest * total *Duration
Change in liabilities	107	
Value change	34	

iii) Company should enter in to receiver swap. The nominal amount can be calculated by backsolving the cash flow of the swap with fixed leg at 10% and floating leg at 7% as given below

	330	10%	7%	
			33.53339	
Swap	payment received		Diff	
1	33.00	23.1	9.90	0.934579
2	33.00	23.1	9.90	0.873439
3	33.00	23.1	9.90	0.816298
4	33.00	23.1	9.90	0.762895

**iv) We should be using the receiver swaption which is designed give option to holder to enter in to fixed to floating swap at a year's duration
Nominal of the swaption – 330 x 80% = 264**

Parameters	
Swaption Start Date	01-Jun-01
Swaption End Date	31-May-04
Swap Start Date	01-Jun-01
Swap Maturity	31-May-04
Number of payments/year	1
Principal Amount	264
1 Year Forward Swap Rate	10.0%

Strike Rate	10.0%
Volatility of Forward Rate	25.0%
Risk Free Rate	10.0%
Results	
Payer	5.9094178
Receiver	5.9094178
Intermediate Calculations	
Swap Tenor (years)	3
Swap Rate Period (years)	1
dt	0.25
d1	0.125
d2	-0.125
nd1	0.5497382
nd2	0.4502618
nnd1	0.4502618
nnd2	0.5497382
A	2.486852

[19 Marks]

Solution 2:

i)

a) Ratchet Caps

Where the cap rate equals the libor rate at previous reset date plus a spread ($K(j+1) = R_j + S$)

b) Sticky Caps

Where the cap rate equals the previous capped rate plus a spread

c) Flexi Caps

Where there is a limit on total number of caplet which can be exercised.

Construction of 1 factor LMM model

ii) Page 25 of interest rate model material

iii) The LMM model would be better suited to price the above options due to the path dependency of the strike price and there is a need to construct the montecarlo simulation

[14.5 Marks]

Solution 3:

i) Gaussian copula

In probability theory and statistics, a copula is a multivariate probability distribution for which the marginal-probability distribution of each variable is uniform. Copulas are used to describe the dependence between random variables. Copulas have been used widely in quantitative finance to model and minimize tail risk and portfolio-optimization applications

The Gaussian copula is a distribution over the unit cube. It is constructed from a multivariate normal distribution over using the probability integral transform.

Steps for simulating the default correction:

- Produce the pseudo random numbers
- Apply Gaussian copula to combine these value with the correct correlation
- Produce a set of correlated values
- Apply percentile to percentile matching to map these values on to the correct default time distribution.
- The produces a set of simulated default time

ii) Performing cholesky decomposition and so on. We choose the coefficients α_{11} so that the correlations and variances are correct. This can be done step by step as follows. Set

$$\alpha_{11} = 1;$$

$$\text{choose } \alpha_{21} \text{ so that } \alpha_{21} \alpha_{11} = \rho_{21};$$

$$\text{choose } \alpha_{22} \text{ so that } \alpha_{21}^2 + \alpha_{22}^2 = 1$$

$$\text{choose } \alpha_{31} \text{ so that } \alpha_{31} \alpha_{11} = \rho_{31};$$

choose

$$\alpha_{32} \text{ so that } \alpha_{31} \alpha_{21} + \alpha_{32} \alpha_{22} = \rho_{32};$$

$$\text{choose } \alpha_{33} \text{ so that } \alpha_{31}^2 + \alpha_{32}^2 + \alpha_{33}^2 = 1;$$

$$\epsilon_1 = \alpha_{11}x_1$$

$$\epsilon_2 = \alpha_{21}x_1 + \alpha_{22}x_2$$

$$\epsilon_3 = \alpha_{31}x_1 + \alpha_{32}x_2 + \alpha_{33}x_3$$

		1	2	3
		A	B	C
1	A	1	0.75	0.3
2	B	0.75	1	0.5
3	C	0.3	0.5	1

alpha11	1
alpha21	0.75
alpha22	0.661438
alpha31	0.3
alpha32	0.415761
alpha33	0.85857
rand 1	0.35
rand 2	0.593219
rand 2	0.870951

iii) Applying the exponential distribution

$$1 - \exp(-\lambda * T)$$

Assuming bonds follow exponential distribution				
Lamda 1	4			1.723132
Lamda 2	8			7.195841
Lamda 3	16			32.76102

[16 Marks]

Solution 4:

- i) Default probability in year 4 is $0.97 * 0.97 * 0.97 * 0.03 = 0.02738019$ or approximately 2.74%
- ii) Probability of survival at the end of 5 years is $0.97 * 0.97 * 0.97 * 0.97 * 0.97 = 0.8587340257$ Or approximately 85.87%
- iii) present value of the payments to be made (both in the case of no default as well as in the accrual payment in the event of default)

Default and Survival Probabilities		
Time (years)	Default Probability	Survival Probability
1	0.0300	0.9700
2	0.0291	0.9409
3	0.0282	0.9127
4	0.0274	0.8853
5	0.0266	0.8587

Calculation of PV of payments; principal = \$1

Time (yrs)	Survival Prob	Expected Paymt	Discount Factor	PV of Exp Pmt
1	0.9700	0.9700s	0.9231	0.8954s
2	0.9409	0.9409s	0.8521	0.8018s
3	0.9127	0.91267s	0.7866	0.7179s
4	0.8853	0.8853s	0.7261	0.6429s
5	0.8587	0.8587s	0.6703	0.5756s
Total				3.6336s

PV of Accrual Payment Made in Event of a Default. (Principal=\$1)

Time	Default Prob	Expected Accr Pmt	Disc Factor	PV of Pmt
0.5	0.0300	0.0150s	0.9608	0.0144s
1.5	0.0291	0.0146s	0.8869	0.0129s
2.5	0.0282	0.0141s	0.8187	0.0116s
3.5	0.0274	0.0137s	0.7558	0.0103s
4.5	0.0266	0.0133s	0.6977	0.0093s
Total				0.0585s

PV of expected payments is $3.6336s + 0.0585s = 3.6921s$

iv) Present Value of the expected payoff in the event of a default

Time (yrs)	Default Probab.	Rec. Rate	Expected Payoff	Discount Factor	PV of Exp. Payoff
0.5	0.03	0.3	0.0210	0.9608	0.0202
1.5	0.0291	0.3	0.0204	0.8869	0.0181
2.5	0.0282	0.3	0.0198	0.8187	0.0162
3.5	0.0274	0.3	0.0192	0.7558	0.0145
4.5	0.0266	0.3	0.0186	0.6977	0.0130
Total					0.0819

PV of expected payment in the event of default is 0.0819

v) Breakeven CDS spread: $3.6921s = 0.0819$ or $s = 0.02218$ or 221.8 bps

vi) Value of a CDS negotiated at an earlier point in time with a CDS spread of 323.98 as a function of the principal:

$$(3.6921 * 0.032398) - 0.0819 = 0.0377 \text{ times the principal}$$

[17.5 Marks]

Solution 5:

- ii) \$85
- iii) \$85.07
- iv) \$21.87

[13 Marks]

Solution 6:

- i) As we increase the frequency, we observe a more extreme minimum, which increases the value of a look back call.
- ii) As we increase the frequency with which the asset price is observed, the asset price becomes more likely to hit the barrier and the value of a down and out call goes down.
- iii) In general, it will be lower since the averaging process reduces overall variance in the risk neutral measure.
- iv) If the value of the sum is higher do vanilla minus the sum; otherwise do the reverse
- v) Enter into a long straddle

[15 Marks]

Solution 7:

FV			1000.000	
Current Price			975.000	
Strike			1000.000	
Option exercise date			11 months	
Interest Rate			0.100	
Volatility			0.100	
Semiannual coupon			50.000	
3 month interest rate			0.095	
9 month interest rate			0.098	
PV of coupons			95.2833	(1 Mark)

Bond Forward Price			964.169	(1 Mark)
d1			-0.33324	(0.5 Mark)
d2			-0.42898	(0.5 Mark)
Option price			20.3193	(2 Marks)

Please note that the PV of coupons is calculated as $[50 \exp(-0.095 \cdot 0.25) + 50 \exp(-0.098 \cdot 0.75)]$ and the bond future price is calculated as $[(\text{Current Bond Price} - \text{PV of coupons}) \exp(0.1 \cdot (11/12))]$ – in this case this is explicitly equal to $[(975 - 95.2833) \exp(0.1 \cdot (11/12))]$.

[5 Marks]
